

Industrial Attachment – Day 4 Report

Synthetic Dataset Preparation & Dataset Study | 8 September 2026 | MySoft Heaven BD

Team: AnomalyX

Team Members

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Date	8 September 2026	Day	Day 4
Company	MySoft Heaven BD	Session	Synthetic Dataset Preparation & Study
Supervised By			
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1. Need for a Bangladesh-Oriented Dataset

On Day 4, we focused on the dataset requirement for our selected project, Enterprise AI Anomaly Detection & Financial Risk Platform. Since our project is designed from a Bangladesh-oriented perspective and focuses on Mobile Financial Services (MFS), we needed data that could represent the type of customers, transactions, devices, and agents relevant to the local context. However, suitable real transaction data from services such as bKash and Nagad was not available to us for the project. Therefore, we decided to create a **synthetic dataset** for our system.

2. Synthetic Dataset Preparation

We started developing a synthetic MFS transaction dataset based on Bangladesh-oriented scenarios. The dataset was designed to represent customers and their transaction activities in an MFS environment similar to the context of services such as bKash and Nagad. Our aim was to create realistic-looking transaction patterns without using actual private customer transaction records.

3. Dataset Structure

The dataset was designed to contain transaction-level information together with customer, device, and agent information. The transaction records include fields such as transaction ID, customer ID, device ID, counterparty, transaction channel, amount, timestamp, location, fraud label, fraud type, and several risk and behavioral indicators. The dataset overview records **392,386 transaction records and 39 columns**. It also represents **5,000 customers, 7,082 devices, and 300 agents**.

4. Fraud and Risk-Related Features

To make the dataset useful for our planned AI-based fraud monitoring system, we included different risk and behavior-related features. These include customer risk score, amount deviation, device risk score, new device indicators, transaction frequency, new counterparty indicators, possible mule patterns, impossible travel indicators, velocity risk score, network risk score, and behavior anomaly score. These features are intended to help the system identify both known fraud patterns and unusual transaction behavior.

5. Fraud Cases and Transaction Patterns

The synthetic dataset includes different fraud categories so that we can study multiple types of suspicious activity. The dataset overview contains categories such as velocity fraud, abnormal cash-out, laundering chain patterns, SIM/device-related fraud, and account takeover. The raw dataset contains **1,177 fraudulent transactions** and **391,209 non-fraudulent transactions**, with an overall fraud rate of approximately **0.300%**.

6. Dataset Preparation for Machine Learning

We also organized the data so that it could later be used for machine learning experiments. The dataset includes separate training, validation, and test sets. The reported ML-ready data contains **274,670 training records**, **58,858 validation records**, and **58,858 test records**. This structure will allow us to train models and evaluate their performance on separate data.

7. Graph-Based Data Preparation

Because our project may also use relationships between customers, agents, devices, and transaction counterparts, the dataset was prepared with graph analysis in mind. The dataset includes graph nodes and edges representing relationships between entities. The overview reports **12,382 graph nodes** and **384,029 graph edges**, which can support future graph-based fraud analysis.

8. Overall Reflection

Day 4 was mainly focused on solving one of the important practical challenges of our project: obtaining suitable data for a Bangladesh-oriented MFS fraud detection system. By creating and studying a synthetic dataset, we established a data foundation for the next stages of our project, including preprocessing, feature preparation, machine learning, anomaly detection, behavioral analysis, and risk scoring.